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REVIEWER ACTIVITIES:
ARMY - AR, MI
AIR FORCE - 84
DLA - IS

USER ACTIVITIES:
ARMY - AV

P. A.: Other Cust OS AT 99

INTERNATIONAL INTEREST

PROCUREMENT SPECIFICATION: MIL-B-5687

SUPERSEDES:

TITLE: BEARING, SLEEVE, FLANGED, SINTERED BRONZE, OIL IMPREGNATED

MILITARY STANDARD

MS 17796

SHEET 1 OF 7

DD FORM 1 MAY 73 672-1 (COORDINATED)

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3120-0592

Ⓐ ENTIRE STANDARD REVISED

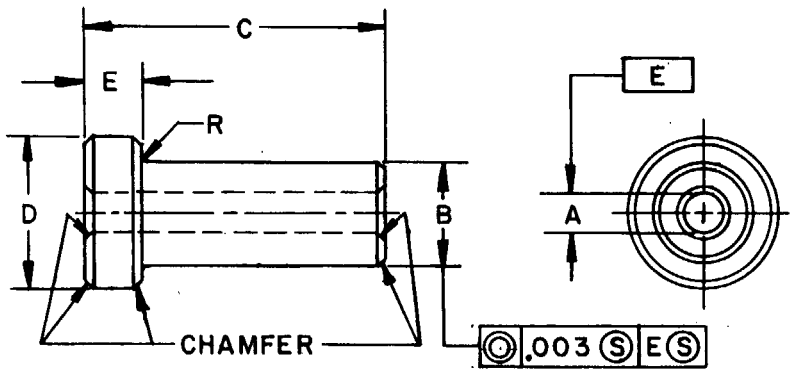


TABLE I. DIMENSIONS AND DASH NUMBERS.

NOMINAL ID	3/32				1/8				5/32		3/16			1/4						
A-BASIC ID	0.095				0.127				0.158		0.1895			0.252						
R-RADIUS (MAX)	1/32				1/32				1/32		1/32			1/32						
B-BASIC OD	0.159				0.1895		0.253		0.253		0.253		0.3155	0.378				0.440		
D-FLANGE OD	3/16		1/4		1/4		3/8		3/8		5/16		7/16	1/2				9/16		
E-FLANGE THK	1/32		1/16		1/32		1/16		1/32		1/16		1/16	1/16		3/64		1/16		
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	
1/8	1	75	—	—	4	96	7	65	—	—	—	—	—	—	—			—	—	
3/16	2	125	136	97	5	160	8	130	140	201	144	195	—	—	—			—	—	
1/4	3	175	137	145	6	222	9	193	141	282	145	290	16	21	387			27	387	
5/16			138	193			10	255	142	363	146	385	17	22	516	—	—	28	516	
3/8			—	—			139	322	143	443	147	485	18	23	645	149	675	29	645	
7/16			—	—			—	—	—	—	148	580	19	24	773	—	—	30	773	
1/2												675	20	25	902			31	902	
5/8												—	—	26	1160			32	1160	
3/4												—	—	—	—			—	—	

APPROVED 16 FEB 1962 REVISED Ⓐ 9 November 1984

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DD FORM 672-1 (COORDINATED)

PROCUREMENT SPECIFICATION: MIL-B-5687
SUPERSEDES:
P. A. Other Cust: OS AT 99
INTERNATIONAL INTEREST:
TITLE: BEARING, SLEEVE, FLANGED, SINTERED
BRONZE, OIL IMPREGNATED
SHEET 2 OF 7
MS 17796
MILITARY STANDARD
FED. SUP CLASS 3120

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TABLE I. DIMENSIONS AND DASH NUMBERS: CONTINUED.

NOMINAL ID	5/16								3/8									
A-BASIC ID	0.3145								0.377									
R-RADIUS (MAX)	1/32				3/64				3/64									
B-BASIC OD	0.440				0.503				0.503							0.628		
D-FLANGE OD	9/16				11/16				5/8			11/16				7/8		
E-FLANGE THK	1/16		3/32		1/16		3/32		1/16		3/32	1/16		3/32		1/16		1/8
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS
1/4	33	485	—	—	—	—	39	405	45	580		—	—	—	—	—	—	—	—
5/16	34	645	—	—	—	—	40	570	46	775		—	—	—	—	—	—	52	580
3/8	35	805	150	725	153	805	41	730	47	965		155	965	158	870	161	965	53	775
13/32	—	—	—	—			—	—	—	965	154	—	—	—	—	—	—	—	—
7/16	36	970	—	—			42	890	48	1160	—	—	—	—	—	—	—	54	965
1/2	37	1130	151	1045			43	1060	49	1355	—	156	1355	159	1255	162	1355	55	1160
5/8	38	1450	—	—			44	1375	50	1740		—	—	—	—	—	—	56	1545
3/4	—	—	152	1690			—	—	51	2125		157	2125	160	2030	163	2125	57	1920

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DD FORM 672-1 (COORDINATED) MAY 73

PROCUREMENT SPECIFICATION: MIL-B-5687
SUPERSEDES:
INTERNATIONAL INTEREST
P. A. OS
Other Cust AT 99

TITLE: BEARING, SLEEVE, FLANGED, SINTERED
BRONZE, OIL IMPREGNATED

MILITARY STANDARD
MS 17796
SHEET 3 OF 7

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TABLE I. DIMENSIONS AND DASH NUMBERS. CONTINUED.

NOMINAL ID	7/16			1/2						5/8				3/4						
A-BASIC ID	0.4395			0.502						0.627				0.752						
R-RADIUS (MAX)	3/64			3/64						3/64				3/64						
B-BASIC OD	0.565			0.628				0.753			0.753		0.879	0.879		0.941			1.004	
D-FLANGE OD	11/16		3/4	3/4		7/8		15/16	1-0		1-0		1-1/8	1-0		1-1/4		1-5/16	1-1/4	
E-FLANGE THK	1/16		1/16	1/16		1/8		1/8	1/8		1/8		1/8	1/16		1/8		1/8	3/16	
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS
3/8	—	1130	58	63	1290	166	1030	—	—	—	—	—	—		—	—	—	—	—	—	—
7/16	—	1360	59	64	1550	167	1290	—	—	—	—	—	—		—	—	—	—	—	—	—
1/2	164	1585	60	65	1810	168	1545	70	1550	173	178	1935	80		183	2705	185	2320	184	194	1935
9/16	—	—	—	—	—	—	—	—	—	—	—	—	—		—	—	—	—	185	—	—
5/8	—	2035	61	66	2325	169	2060	71	2070	174	179	2580	81		—	—	—	—	—	—	—
3/4	165	2490	62	67	2840	170	2580	72	2580	175	180	3220	82		184	4255	186	3865	186	195	3480
7/8				68	3355	171	3095	73	3100	176	—	—	83	3865			—	—	—	—	—
1-0				—	—	172	3610	74	3615	177	181	4510	84	—			187	5415	187	196	5025
1-1/8				—	—	—	—	—	—	—	182	5155	85	—			—	—	—	—	—
1-1/4				—	—	—	—	—	—	—	—	—	86	5800			188	6960	188	197	6575

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USER ACTIVITIES:
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TABLE I. DIMENSIONS AND DASH NUMBERS. CONTINUED.

NOMINAL ID	7/8				1-0						1-1/4					1-1/2					
A-BASIC ID	0.877				1.003						1.2535					1.504					
R-RADIUS (MAX)	3/64				3/64						1/16		3/64		3/64	1/16			3/64		3/64
B-BASIC OD	1.004		1.129		1.254			1.379			1.504		1.504		1.630	1.755			1.755		1.880
D-FLANGE OD	1-1/4		1-1/2		1-1/2		1-7/8	1-5/8		1-3/4	1-3/4		1-3/4		1-3/4	2-0			2-0		2-0
E-FLANGE THK	3/16		1/8		1/8		1/8	3/16		3/16	3/16		3/16		3/16	1/8			3/16		3/16
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP	DASH NO	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS
5/8	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
3/4	198	4060	200	4510	205	5155	—	210	4640	117	—	—	—	—	—						
1-0	199	5865	201	6315	206	7220	209	211	6705	118	123	8380	215	219	8380						
1-1/4			202	8120	207	9280		—	8760	119	124	10960	216	—	—	—	—	—	—	—	—
1-1/2			203	9925	208	11345		212	10730	120	125	13540	217	220	13540	129	17015	221	13150	222	13150
1-3/4			204	11730	—	—		213	12895	121	126	16121	218	—	—	—	—	—	—	—	—
2-0								214	14955	122	127	18695				130	23200			—	—
2-1/4								—	—	—	—	—				—	—			223	25525
2-1/2								—	—	—	128	23850				131	29390			—	—

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USER ACTIVITIES:
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TABLE I. DIMENSIONS AND DASH NUMBERS. CONTINUED.

NOMINAL ID	1-3/4		2-0				2-1/2				3-0	
A-BASIC ID	1.754		2.004				2.505				3.006	
R-RADIUS (MAX)	3/32		1/16				3/32				3/32	
B-BASIC OD	2.254		2.254		2.505		3.006				3.507	
D-FLANGE OD	3-0		2-1/2		3-0		3-1/4		3-1/2		3-3/4	
E-FLANGE THK	1/4		1/8		1/4		1/8		1/4		1/4	
C LENGTH	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS	DASH NO	STAT CAP LBS
1-3/4	132	21660	—	—	—	—	—	—	—	—	—	—
2-0	133	25270	224	30935	226	28875	228	38675	231	36095	—	—
2-1/2	134	32485	—	—	—	—	—	—	—	—	234	55685
3-0	135	39705	225	47440	227	45375	229	59295	232	56720	235	68060
3-1/2	—	—	—	—	—	—	—	—	—	—	236	80435
4-0	—	—	—	—	—	—	230	79920	233	77345	237	92810
4-1/2											238	105185
5-0											239	117560

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P.A. OS Other Cust AT 99
INTERNATIONAL INTEREST
SUPERSEDES:
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TITLE BEARING, SLEEVE, FLANGED, SINTERED BRONZE, OIL IMPREGNATED
MILITARY STANDARD
MS 17796
SHEET 5 OF 7

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DD FORM 672-1 MAY 73 (COORDINATED)

PROCUREMENT SPECIFICATION MIL-B-5687

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TITLE
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SHEET 6 OF 7

MILITARY STANDARD
MS 17796

FED. SUP CLASS 3120

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TABLE II. DIMENSIONAL TOLERANCES.

DIMENSION	BEARING SIZE	TOLERANCE
INSIDE DIAMETER "A" AND OUTSIDE DIAMETER "B"	UP TO 1.510 ID INCL	+.000 -.001
	1.511 TO 2.510	+.0000 -.0015
	2.511 TO 3.010	+.000 -.002
	3.011 TO 3.507	+.000 -.003
FLANGE OUTSIDE DIAMETER "D"	UP TO 1.510 FLANGED OD INCL.	±0.005
	1.511 TO 3.010	±0.010
	3.011 TO 3.750	±0.025
FLANGE THICKNESS "E"	UP TO 1.510 FLANGED OD	± .005
	1.511 TO 3.010 FLANGED OD	± .010
	3.011 TO 3.750 FLANGED OD	± .015
LENGTH "C"	UP TO 1.495 INCL	±0.005
	1.496 TO 2.990	±0.0075
	2.991 TO 4.985	±0.010
CONCENTRICITY FULL INDICATOR MOVEMENT (FIM)	OUTSIDE DIAMETER (BASIC)	TOLERANCES
	UP TO 1.510 1.511 TO 2.010 2.011 TO 3.507	0.003 0.004 0.005

TABLE III. CHAMFERS.

RANGE	SIZE
WALL THICKNESS UP TO AND INCLUDING 3/32"	BREAK EDGES
WALL THICKNESS GREATER THAN 3/32" UP TO 3" OUTSIDE DIAMETER	1/64 INCH x45 DEGREES
ALL BEARINGS 3" OUTSIDE DIAMETER AND OVER	1/32 INCH x45 DEGREES

TABLE IV. SHAFT RUNNING CLEARANCE.

SHAFT SIZE	CLEARANCE MIN
UP TO 0.760	0.0005
0.761 TO 1.510	0.0010
1.511 TO 2.510	0.0015
OVER 2.510	0.0020

TABLE V. RECOMMENDED PRESS FITS.

OUTSIDE DIAMETER "B"	PRESS FITS	
	MAX	MIN
UP TO 0.760	0.003	0.001
0.760 TO 1.510	0.004	0.0015
1.511 TO 2.510	0.005	0.002
2.511 TO 3.010	0.006	0.002
OVER 3.010	0.007	0.002

TABLE VI. PERMISSIBLE LOAD RATINGS.

SHAFT VELOCITY (FEET PER MINUTE)	PERMISSIBLE LOAD (POUNDS PER SQUARE INCH)
SLOW AND INTERMITTENT	4000
25	2000
26 TO 50 INCL	1000
51 TO 100 INCL	500
101 TO 150 INCL	325
151 TO 200 INCL	250
OVER 200	SEE REQUIREMENT 9

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DD FORM 672-1 1 MAY 73 (COORDINATED)

PROCUREMENT SPECIFICATION: MIL-B-5687

P. A. OS AT 99 Other Cust

INTERNATIONAL INTEREST

SUPERSEDES:

TITLE: BEARING, SLEEVE, FLANGED, SINTERED BRONZE, OIL IMPREGNATED

MILITARY STANDARD

MS 17796

SHEET 7 OF 7

FED. SUP CLASS 3120

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REQUIREMENTS:

1. THESE BEARINGS ARE NOT INTENDED FOR REAMING ON ASSEMBLY.
2. MATERIAL: SINTERED, COPPER BASE, POWDERED METAL IN ACCORDANCE WITH MIL-B-5687. TYPE I, GRADE 1 OR 2. YIELD STRENGTH IN COMPRESSION, 11,000 PSI (MIN) FOR 0.1% PERMANENT SET.
3. HARDNESS: 30-70 HRH.
4. POROSITY: (OIL CONTENT BY VOLUME) 20% MINIMUM.
5. LUBRICATION: BEARINGS ARE TO BE IMPREGNATED 18% BY VOLUME WITH OIL CONFORMING TO MIL-L-17331.
6. DIMENSIONS: ALL DIMENSIONS ARE IN INCHES.
7. DIMENSIONAL TOLERANCES AND CHAMFERS. SEE TABLES II, III, IV, AND V.
8. LOAD RATINGS: SEE TABLE VI AND REQUIREMENT 9.
9. PERMISSIBLE LOAD: FOR SHAFT VELOCITIES ABOVE 200 FEET PER MINUTE, THE PERMISSIBLE LOAD IS FOUND FROM THE FOLLOWING;

$$P = \frac{50{,}000}{V}$$ WHERE P = SAFE LOAD, PSI OF PROJECTED AREA (π x BEARING ID x LENGTH)
V = SHAFT VELOCITY IN FT/MIN

10. STATIC CAPACITY: SPECIFIC STATIC CAPACITY IS 75% OF THE MAXIMUM PURE RADIAL LOAD WHICH WILL PRODUCE A PERMANENT SET OF LESS THAN 0.1% IN BEARING AT ANY CONTACT AREA BETWEEN SHAFT AND BEARING AT ZERO VELOCITY. (NOTE: THE SPECIFIC STATIC CAPACITY GIVEN IS INTENDED AS A GUIDE IN SELECTING THE PROPER BEARING AND NO TESTS ARE REQUIRED TO DETERMINE THIS FACTOR FOR ACCEPTANCE OF BEARING.)
11. STATIC LOAD: THE MAXIMUM LOAD SHOULD NOT EXCEED 8,500 PSI OF PROJECTED BEARING AREA (LENGTH x INSIDE DIAMETER OF BEARING).
12. RADIAL CRUSHING STRENGTH: THE RADIAL CRUSHING STRENGTH OF CYLINDRICAL PARTS SHALL BE NOT LESS THAN THE VALUE DETERMINED FROM THE FOLLOWING;

$$S = \frac{Lt^2 x 22{,}500}{D-t}$$ WHERE S = RADIAL CRUSHING STRENGTH IN POUNDS
D = OD OF BEARING IN INCHES
t = WALL THICKNESS IN INCHES
L = LENGTH OF BEARING IN INCHES

13. PART NUMBER. THE BASIC MILITARY STANDARD (MS) PART NUMBER IS FOLLOWED BY A DASH NUMBER SELECTED FROM TABLE I.

EXAMPLE: MS17796-1
- DASH NUMBER FROM TABLE I
- BASIC MS NUMBER

NOTES:

1. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
2. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSALS, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
3. NOT RECOMMENDED FOR MILITARY AIRFRAME APPLICATIONS.

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